

Q1 - 24 June - Shift 1

In an examination, there are 5 multiple choice questions with 3 choices, out of which exactly one is correct. There are 3 marks for each correct answer, -2 marks for each wrong answer and 0 mark if the question is not attempted. Then, the number of ways a student appearing in the examination gets 5 marks is _.

*Space for your notes:***Q2 - 24 June - Shift 2**

The number of 7-digit numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is _____.

*Space for your notes:***Q3 - 24 June - Shift 2**

The sum of all the elements of the set $\{\alpha \in \{1, 2, \dots, 100\} : \text{HCF}(\alpha, 24) = 1\}$ is _____.

*Space for your notes:***Q4 - 25 June - Shift 1**

The number of 3-digit odd numbers, whose sum of digits is a multiple of 7, is _____.

*Space for your notes:***Q5 - 25 June - Shift 2**

The total number of three-digit numbers, with one digit repeated exactly two times, is _____.

*Space for your notes:***Q6 - 26 June - Shift 1**

Questions

MathonGo

There are ten boys $B_1, B_2, \dots, B_{10}$ and five girls $G_1, G_2, \dots, G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the members of a group, is _____.

*Space for your notes:***Q7 - 26 June - Shift 2**

The total number of 3-digit numbers, whose greatest common divisor with 36 is 2, is _____.

*Space for your notes:***Q8 - 27 June - Shift 1**

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in a row so that between any two red cubes there should be at least 2 blue cubes, is _____.

*Space for your notes:***Q9 - 28 June - Shift 1**

The total number of 5-digit numbers, formed by using the digits 1, 2, 3, 5, 6, 7 without repetition, which are multiple of 6, is

Space for your notes:

- (A) 36 (B) 48
(C) 60 (D) 72

Q10 - 28 June - Shift 1

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Questions

MathonGo

Let $A = \{1, a_1, a_2, \dots, a_{18}, 77\}$ be a set of integers with $1 < a_1 < a_2 < \dots < a_{18} < 77$. Let the set $A + A = \{x + y : x, y \in A\}$ contain exactly 39 elements. Then, the value of $a_1 + a_2 + \dots + a_{18}$ is equal to _____.

Space for your notes:

Q11 - 28 June - Shift 2

The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3 and C_4 so that C_2 receives atleast 4 and atmost 7 candies, C_3 receives atleast 2 and atmost 6 candies, is equal to

- (A) 205 (B) 615
(C) 510 (D) 430

Space for your notes:

Q12 - 29 June - Shift 1

Let $b_1 b_2 b_3 b_4$ be a 4-element permutation with $b_i \in \{1, 2, 3, \dots, 100\}$ for $1 \leq i \leq 4$ and $b_i \neq b_j$ for $i \neq j$, such that either b_1, b_2, b_3 are consecutive integers or b_2, b_3, b_4 are consecutive integers.

Then the number of such permutations $b_1 b_2 b_3 b_4$ is equal to _____.

Space for your notes:

Q13 - 29 June - Shift 2

The total number of four digit numbers such that each of the first three digits is divisible by the last digit, is equal to _____.

Space for your notes:

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Answer Key**Q1** (40)**Q2** (576)**Q3** (1633)**Q4** (63)**Q5** (243)**Q6** (1120)**Q7** (150)**Q8** (56)**Q9** (D)**Q10** (702)**Q11** (D)**Q12** (18915)**Q13** (1086)

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Q1 (40)

$$x_1 + x_2 + x_3 + x_4 + x_5 = 5$$

Only one possibilities 3, 3, 3, -2, -2

$$\text{Number of ways is} = \frac{5!}{3!2!} \times 2 \times 2 = 40$$

Q2 (576)

Digits are 1, 2, 3, 4, 5, 7, 9

Multiple of 11 $\rightarrow$ Difference of sum at even & odd place is divisible by 11.

Let number of the form abcdefg

$$\therefore (a + c + e + g) - (b + d + f) = 11x$$

$$a + b + c + d + e + f = 31$$

$$\therefore \text{either } a + c + e + g = 21 \text{ or } 10$$

$$\therefore b + d + f = 10 \text{ or } 21$$

Case- 1

$$a + c + e + g = 21$$

$$b + d + f = 10$$

$$(b, d, f) \in \{(1, 2, 7), (2, 3, 5), (1, 4, 5)\}$$

$$(a, c, e, g) \in \{(1, 4, 7, 9), (3, 4, 5, 9), (2, 3, 7, 9)\}$$

$$\therefore \text{Total number in case-1} = (3! \times 3) (4!) = 432$$

Case- 2

$$a + c + e + g = 10$$

$$b + d + f = 21$$

$$(a, b, e, g) \in \{1, 2, 3, 4\}$$

$$(b, d, f) \in \{(5, 7, 9)\}$$

$$\therefore \text{Total number in case 2} = 3! \times 4! = 144$$

$$\therefore \text{Total numbers} = 144 + 432 = 576$$

Q3 (1633)

Hints and Solutions

MathonGo

$$\text{HCF}(\alpha, 24) = 1$$

$$\text{Now, } 24 = 2^2 \cdot 3$$

→ α is not the multiple of 2 or 3

Sum of values of α

$$= S(U) - \{S(\text{multiple of } 2) + S(\text{multiple of } 3)$$

$$- S(\text{multiple of } 6)\}$$

$$= (1 + 2 + 3 + \dots + 100) - (2 + 4 + 6 + \dots + 100) - (3$$

$$+ 6 + \dots + 99) + (6 + 12 + \dots + 96)$$

$$= \frac{100 \times 101}{2} - 50 \times 51 - \frac{33}{2} \times (3 + 99) + \frac{16}{2} (6 + 96)$$

$$= 5050 - 2550 - 1683 + 816 = 1633 \text{ Ans.}$$

Q4 (63)

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To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

Hints and Solutions

MathonGo

$x, y, z \leftarrow \text{odd number}$ $z = 1, 3, 5, 7, 9$ $x+y+z = 7, 14, 21$ [sum of digit multiple of 7] $\begin{matrix} x & y \\ 1 \text{ to } 9 & 0 \text{ to } 9 \end{matrix} = 6, 4, 2, 13, 11, 9, 7, 5, 20, 18, 16, 14, 12$					
$x + y = 6 \Rightarrow (1,5), (2, 4), (3, 3), (4, 2), (5, 1), (6, 0)$ $\rightarrow \text{T.N.} = 6$					
$x + y = 4 \Rightarrow (1,3), (2, 2), (3, 1), (4,0)$ $\rightarrow \text{T.N.} = 4$					
$x + y = 2 \Rightarrow (1,1), (2,0)$ $\rightarrow \text{T.N.} = 2$					
$x + y = 13 \Rightarrow (4,9), (5,8), (6,7), (7,6), (8,5), (9,4)$ $\rightarrow \text{T.N.} = 6$					
$x + y = 11 \Rightarrow (2,9), (3,8), (4,7), (5,6), (6,5), (6,5), (7,4), (8,3), (9,2)$ $\rightarrow \text{T.N.} = 8$					
$x + y = 9 \Rightarrow (1,8), (2,7), (3,6), (4,5), (5,4), \dots, (8,1), (9,0)$ $\rightarrow \text{T.N.} = 9$					
$x + y = 7 \Rightarrow (1,6), (2,5), (3,4), \dots, (6, 1), (7,0)$ $\rightarrow \text{T.N.} = 7$					
$x + y = 5 \Rightarrow (1,4), (2,3), (3, 2), (4,1), (5,0)$ $\rightarrow \text{T.N.} = 5$					
$x + y = 20 \Rightarrow$ Not possible					
$x + y = 18 \Rightarrow (9,9)$ $\rightarrow \text{T.N.} = 1$					
$x + y = 16 \Rightarrow (7,9), (8,8), (9,7)$ $\rightarrow \text{T.N.} = 3$					
$x + y = 14 \Rightarrow (5,9), (6,8), (7,7), (8,6), (9,5)$ $\rightarrow \text{T.N.} = 5$					
$x + y = 12 \Rightarrow (3,9), (4,8), (5,7), (6,6), \dots, (9,3)$ $\rightarrow \text{T.N.} = 7$					

Q5 (243)

If 0 taken twice then ways = 9

If 0 taken once then ${}^9C_1 \times 2 = 18$

If 0 not taken then ${}^9C_1 {}^8C_1 \cdot 3 = 216$

Total = 243

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Q6 (1120)

$$n(B) = 10$$

$$n(a) = 5$$

The number of ways of forming a group of 3 girls of 3 boys.

$$= {}^{10}C_3 \times {}^5C_3$$

$$= \frac{10 \times 9 \times 8}{3 \times 2} \times \frac{5 \times 4}{2} = 1200$$

The number of ways when two particular boys B_1

of B_2 be the member of group together

$$= {}^8C_1 \times {}^5C_3 = 8 \times 10 = 80$$

Number of ways when boys B_1 of B_2 not in the same group together

$$= 1200 \times 80 = 1120$$

Q7 (150)

$$36 = 2 \times 2 \times 3 \times 3$$

Number should be odd multiple of 2 and does not having factor 3 and 9

Odd multiple of 2 are

102, 106, 110, 114 998 (225 no.)

No. of multiples of 3 are

102, 114, 126 990 (75 no.)

Which are also included multiple of 9

Hence,

$$\text{Required} = 225 - 75 = 150$$

Q8 (56)

16 cubes $\begin{cases} 11 \text{ Blue} \\ 5 \text{ Red} \end{cases}$

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 11$$

$$x_1, x_6 \geq 0, \quad x_2, x_3, x_4, x_5 \geq 2$$

$$x_2 = t_1 + 2$$

$$x_3 = t_3 + 2$$

$$x_4 = t_4 + 2$$

$$x_5 = t_5 + 2$$

$$x_1, t_2, t_3, t_4, t_5, x_6 \geq 0$$

$$\text{No. of solutions} = {}^{6+3-1}C_3 = {}^8C_3 = 56$$

Q9 (D)

To make a no. divisible by 3 we can use the digits

1,2,5,6,7 or 1,2,3,5,7.

Using 1,2,5,6,7, number of even numbers is

$$= 4 \times 3 \times 2 \times 1 \times 2 = 48$$

Using 1,2,3,5,7, number of even numbers is

$$= 4 \times 3 \times 2 \times 1 \times 1 = 24$$

Required answer is 72.

Q10 (702)

$a_1, a_2, a_3, \dots, a_{18}, 77$

are in AP i.e. 1, 5, 9, 13, ..., 77.

Hence $a_1 + a_2 + a_3 + \dots + a_{18} = 5+9+13+\dots$ 18 terms

$$= 702$$

Q11 (D)

$$t_1 + t_2 + t_3 + t_4 = 30$$

$$\text{Coefficient of } x^{30} \text{ in } (1 + x + x^2 + \dots + x^{30})^2$$

$$(x^4 + x^5 + x^6 + x^7) (x^2 + x^3 + x^4 + x^5 + x^6)$$

$$x^6 \left(\frac{1-x^{31}}{1-x} \right)^2 (1+x+x^2+x^3)(1+x+x^2+x^3+x^4)$$

$$x^6 (1-x^3)^2 (1-x^4) (1-x^5) (1-x)^4$$

$$x^6 (1-x^4-x^5+x^9) (1+x^{62}-2x^{31}(1-x)^{-4})$$

$$\text{Coefficient of } x^n \text{ in } (1-x)^{-r} \text{ is } {}^{n+r-1}C_{r-1}$$

$$\Rightarrow {}^{27}C_3 - {}^{23}C_3 - {}^{22}C_3 + {}^{18}C_3$$

$$= 2925 - 1771 - 1540 + 816$$

$$= 430$$

OR

$$x_2 \in [4, 7], x_3 \in [2, 6]$$

$$\Rightarrow t_1 + t_2 + t_3 + t_4 = 24$$

$$\text{total ways} =$$

$${}^{24+4-1}C_{4-1} - {}^{20+4-1}C_{4-1} - {}^{19+4-1}C_{4-1} + {}^{15+4-1}C_{4-1}$$

$$= {}^{27}C_3 - {}^{23}C_3 - {}^{22}C_3 + {}^{18}C_3 = 430$$

Q12 (18915)

$$b_i \in \{1, 2, 3, \dots, 100\}$$

Let A = set when b_1, b_2, b_3 are consecutive

$$n(A) = \frac{97 + 97 + \dots + 97}{98 \text{ times}} = 97 \times 98$$

Similarly when b_2, b_3, b_4 are consecutive

$$N(A) = 97 \times 98$$

$$n(A \cap B) = \frac{97 + 97 + \dots + 97}{98 \text{ times}} = 97 \times 98$$

Similarly when b_2, b_3, b_4 are consecutive

$$n(B) = 97 \times 98$$

$$n(A \cap B) = 97$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\text{Number of permutation} = 18915$$

Q13 (1086)

Let the number is $abcd$, where a, b, c are divisible by d .

No. of such numbers

$$d = 1, \quad 9 \times 10 \times 10 = 900$$

$$d = 2, \quad 4 \times 5 \times 5 = 100$$

$$d = 3, \quad 3 \times 4 \times 4 = 48$$

$$d = 4, \quad 2 \times 3 \times 3 = 18$$

$$d = 5, \quad 1 \times 2 \times 2 = 4$$

$$d = 6, 7, 8, 9, \quad 4 \times 4 = 16$$

1086

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